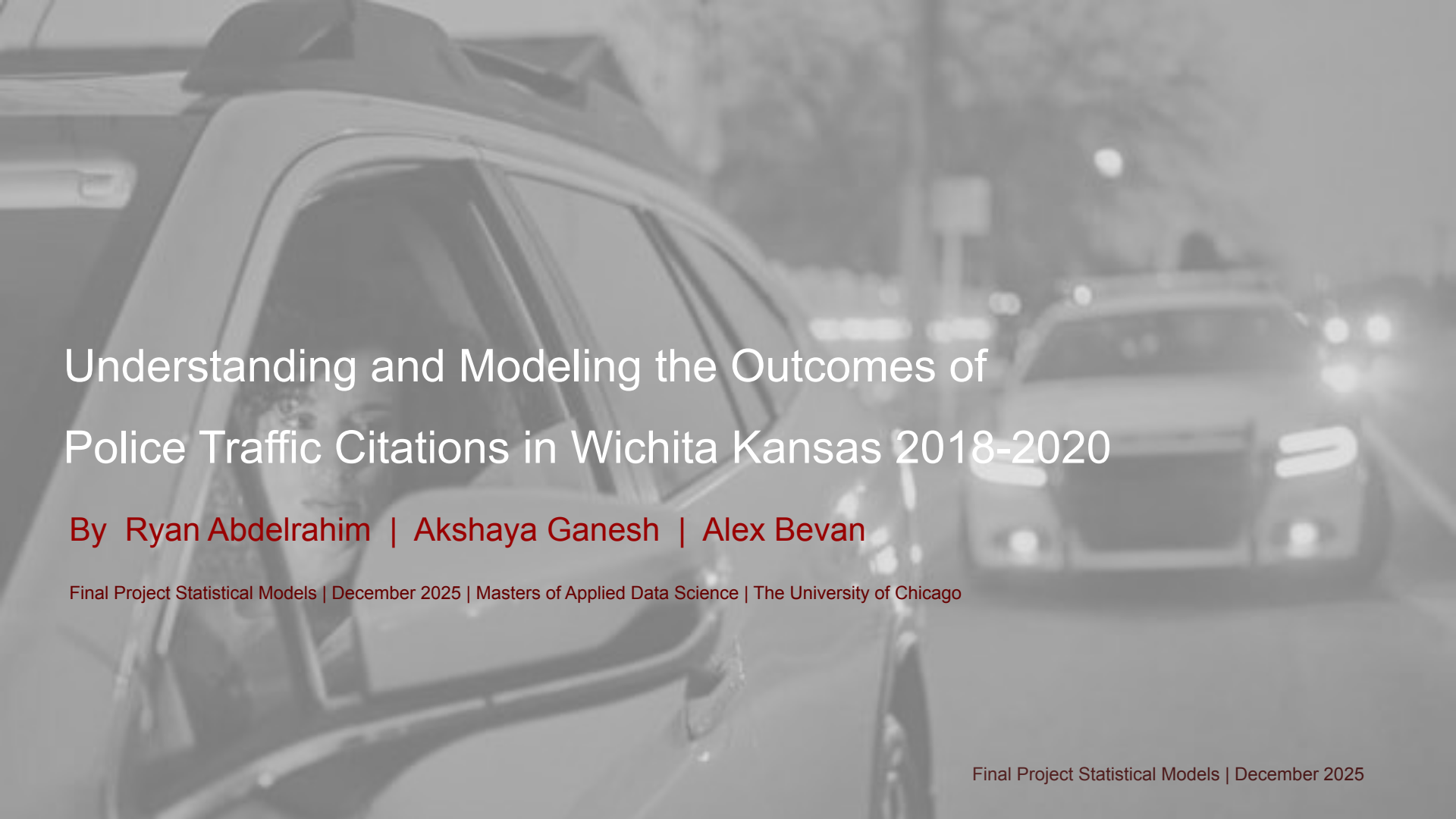




Understanding and Modeling the Outcomes of Police Traffic Citations in Wichita Kansas 2018-2020

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Final Project Statistical Models | December 2025 | Masters of Applied Data Science | The University of Chicago

Motivations, Dataset, Research Question

- **The Stanford Open Policing Project**

E. Pierson, C. Simoiu, J. Overgoor, S. Corbett-Davies, D. Jenson, A. Shoemaker, V. Ramachandran, P. Barghouty, C. Phillips, R. Shroff, and S. Goel. "A large-scale analysis of racial disparities in police stops across the United States". Nature Human Behaviour, Vol. 4, 2020.

- **Wichita, Kansas:** Population 398,000 in 2018. 63.5% is white. Data on 1,030,376 stops dated Dec 2005 to May 2020, including disposition data (official outcome of the citation)
- **Research Question:** What is the relationship of the recorded factors affiliated with traffic citations in Wichita between 2018 and 2020 (i.e., race, location, vehicle make) to the formal outcomes of those citations?

Necessary Context

- **Public Scrutiny:** This is all **pre-George Floyd (May 25 2020)**. Prior to 2018, Wichita Police Department had been under **heavy public scrutiny** for racial discrimination prompting **changes to use of force tactics** in 2018. Between 2013 and 2023, Wichita police showed a 92% in racial disparities involving deadly force (policescorecard.org uses federal and state policing databases)
- **Hispanic Drivers:** Hispanics make of 18.4% of Wichita. Beginning in 2018, the ethnic category of “**hispanic**” in police citations was discontinued and collapsed into “unknown” and “other”.
- **Recording Race and Sex:** During a police stop, race is **assigned by the officer based on visual assessment**
- The **subject sex level “unknown”** is a blackbox: It could be queer publics or it could just be documentation error or a combo of the two.

Variable	Count	Percentage From Whole Dataset
Vehicle year	747488	72.5
Posted speed	709075	68.8
Vehicle model (i.e. Fitt)	519946	50.5
Raw defendant ethnicity	389246	37.8
Subject sex	219581	21.3
Subject age	191387	18.6
Subject race	67551	6.6
Raw defendant race	67551	6.6
Vehicle color	60034	4.5
Vehicle make (i.e. Honda)	46367	4.5
Disposition (i.e. Guilty)	36616	3.6
Geocode source	27487	2.7
Lat and Long	27487	2.7
Location	26738	2.6

Variables & Missing Data Management

Red : Columns dropped

Gray: Reassigned to different bins or dropped those rows entirely

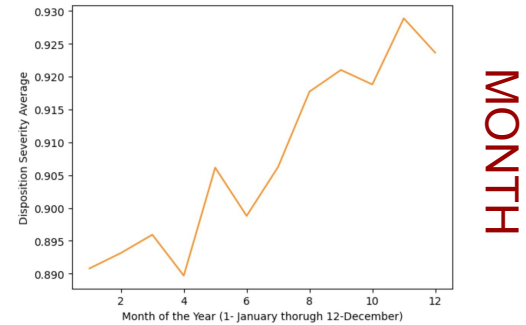
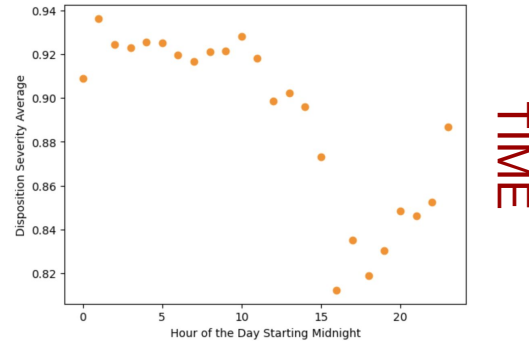
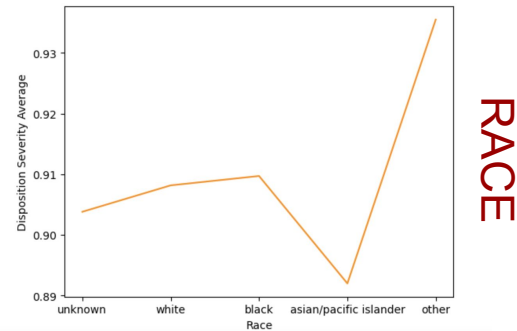
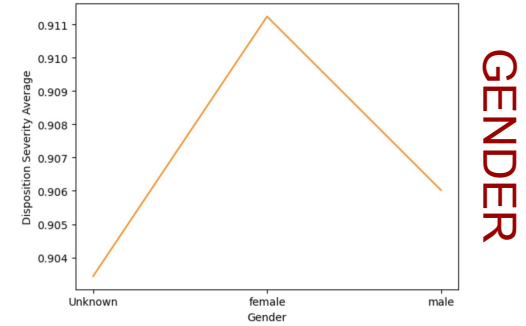
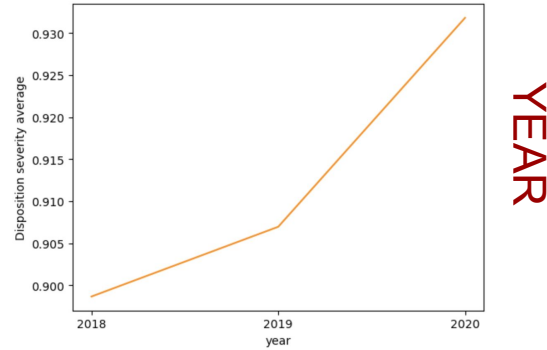
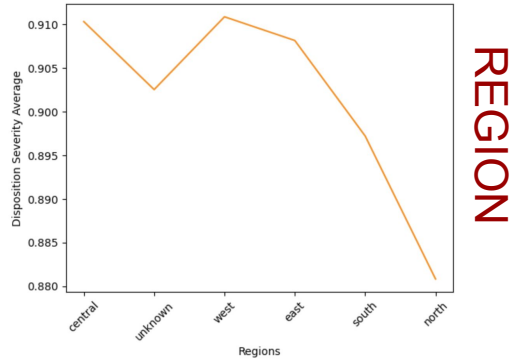
Disposition is our target variable.
Rows with missing “disposition” were dropped

Cleaned dataset is 86,830 from 1 mil



Data Description

Key Variables Against Mean Disposition Severity



Most Severe Outcomes by Race

Other/Unknown
Disposition Severe

Black
Disposition Severe

White
Disposition Severe

Asian/Pacific Isl.
Disposition Severe



Model Selection

Model Selection

Model Type: Binomial regression

Estimation Method: Maximum Likelihood Estimation (MLE)

Target Variable: Disposition severity divided into more severe (1) and less severe (0)

- Disposition 1 examples: “Guilty”, “Nollo Contendre”
- Disposition 0 examples: “Amended Charge”, “Not Guilty”



Variable Selection

Criteria

Three criteria used to determine whether to keep or drop a variable for the model:

- **Statistical diagnostics**
- **Domain relevance**
- **Predictive contribution**

Exclusions made if a variable failed one or more criteria

Variable Group - City Location

Split into dummies - **North, South, East, West, Central** and **Unknown**

Statistical Soundness: No multicollinearity amongst dummies.

Domain Relevance: Enforcement patterns differ across Wichita's geography (affluent east vs. historically Black N/NE etc.). ^{[1], [2]}

Predictive Contribution: Each dummy improved explanatory power

Decision: Retained all except baseline (Central)

Variable Group - Subject Race

Split into dummies - **White, Black, Asian/Pacific Islander, Other** and **Unknown**

Statistical Soundness: VIF check flagged “unknown” (~70) so it is excluded.

Domain Relevance: Race, recorded based on officer discretion, is central to disparities in traffic enforcements. [3],[4],[5], [16] *Reminder: Hispanic was no longer reported post 2018 due to policy changes.*

Predictive Contribution: Each dummy improved explanatory power.

Decision: Retained all except baseline (White) and unknown

Variable Group - Vehicle Tier

Split into dummies - **Luxury, Economy, Mid, Other** (legacy vehicles, motorcycles, trucks, trailers and RVs) and **Unknown**

Statistical Soundness: No multicollinearity amongst dummies.

Domain Relevance: Vehicle tiers show vast difference in citation counts and disposition outcomes [6], [7], [8], [9], [10], [11], [12]

- Luxury vehicle drivers with “veil of affluence”
- Subcategories in “other” have stricter enforcements

Predictive Contribution: Each dummy improved explanatory power.

Decision: Retained all except baseline (Economy)

Variable Group - Age Categories

Split into dummies - ages **16-24, 25-34, 35-54, 55-64, 65+**

Statistical Soundness: No multicollinearity amongst dummies.

Domain Relevance: Disposition severity decreases with age as drivers move away from the highest-risk “young driver” phase (16–24). [16], [17], [18], [19], [20], [21]

Predictive Contribution: Each dummy improved explanatory power. 65+ had a coefficient very close to zero and was insignificant.

Decision: Retained all except baseline (16-24) and 65+.

Variable Group - Date and Time

Geometrical transformations (sin and cosine) were applied for citation **month, day of the week and hour**

Statistical Soundness: No multicollinearity amongst dummies.

Domain Relevance: Citations attached to harsher outcomes occur at/during:

- Later hours ^[25]
- Winter months (holiday drinking and road conditions)^{[22][24]}
- Weekends (DUIs, nightlife traffic and late night driving risks)^{[23][24]}

Predictive Contribution: Each dummy improved explanatory power

Decision: Retained all

Variable Groups - Subject Sex, Citation Year, and Violation Severity

Turn into dummies:

- **2018, 2019 and 2020**
- **Violation severity:** 1, 2, and 3
- **Male, Female** and **Unknown**

Statistical Soundness: Sex Unknown had a high VIF (~70)

Domain Relevance:

- Stricter enforcements happen over time [26], [27]
- Higher violation severity tends to lead to harsher disposition [28]
- Males tend to be stopped more often than females [16]

Predictive Contribution: Each dummy improved explanatory power except for sex male (coefficient very close to zero and insignificant)

Decision: Retained all except baselines (year 2020, violation category 1) and all of subject sex

Variable Strategy Overview

- In total, **26 variables** were retained, each meeting at least two of the three criteria.
- **Exclusions** were made only when statistical diagnostics indicated redundancy or instability
- **Interaction terms** were attempted for variables but all either provided no new helpful information for the model or made the model worse
- Events per variable = $86,830/26 \sim 3340$ **observations per variable**
- **Skewness** was handled with the geometric transformations



Results & Interpretation

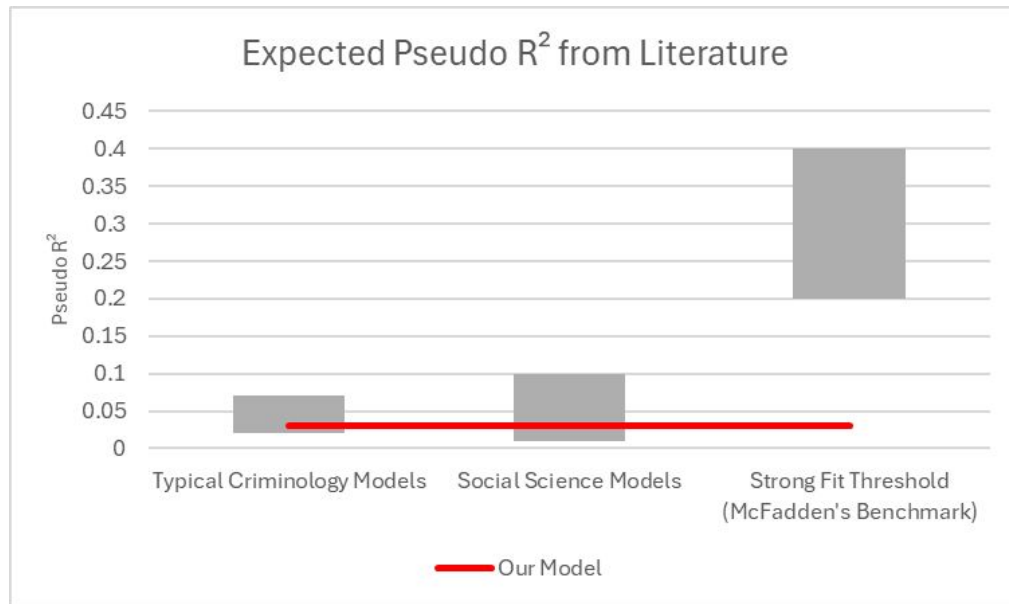
Model Fit Overview

Key Metrics:

- AIC = **51628.258**
- Log - Likelihood: **-25,787.129**
- Deviance: **51,574**
- Pseudo R^2 (Cox & Snell): **0.03046**

Interpretation:

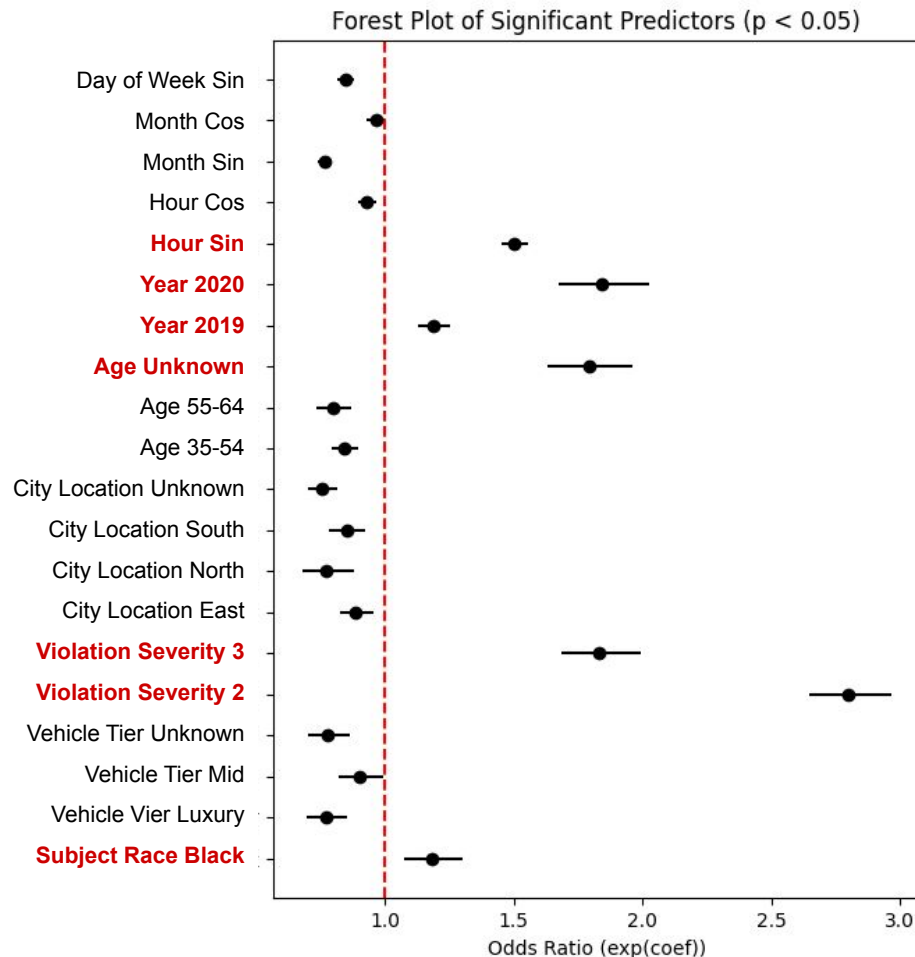
Model explains ~3% of variance which is small but falls within the **expected range (0.02 - 0.07)** for criminology models due to the complexity of human behavior



Expected pseudo R^2 values from literature vs our model [29], [30], [31]

Significant Predictors and Intercept

- Intercept value is 1.7386 which means our **baseline probability for severe disposition is 85%**
- **The forest plot** shows the odds ratio of significant variables where odds ratio greater than 1 increases the odds of a severe disposition and less than 1 decreases it



Significance & Takeaways

The table presents the chosen variables, their coefficients and p-values.

Gray- not significant (p-value > 0.05)

White - borderline (0.05 - 0.1)

Yellow - significant (p<0.05)

Strongest drivers:

- **Violation severity**
- **Year**
- **Age unknown**
- **Race (Black)**
- **Time of day** (max prob at 6:41 AM and min at 6:41 PM)

Variable	Coefficient	Odds Ratios	pval
Subject Race Black	0.167063	1.1818	0.0007
Subject Race Asian	-0.177142	0.8377	0.0530
Subject Race Other	0.494069	1.6394	0.1839
Vehicle Tier Luxury	-0.262132	0.7695	0.0000
Vehicle Tier Mid	-0.103824	0.9015	0.0298
Vehicle Tier Unknown	-0.205110	0.8146	0.0001
Violation Severity 2	1.029977	2.8009	0.0000
Violation Severity 3	0.605495	1.832	0.0000
City Location East	-0.120248	0.8867	0.0001
City Location North	-0.258910	0.7718	0.0001
City Location South	-0.162309	0.8502	0.0001
City Location Unknown	-0.282344	0.7539	0.0000
City Location West	-0.054007	0.9474	0.2170
Age 24-34	-0.039230	0.9615	0.2502
Age 35-54	-0.173047	0.8412	0.0001
Age 55-64	-0.227732	0.7963	0.0000
Age Unknown	0.582325	1.7902	0.0000
Year 2019	0.173959	1.1898	0.0000
Year 2020	0.609676	1.8399	0.0000
Hour Sin	0.043940	1.0449	0.0001
Hour Cos	-0.073205	0.9294	0.0002
Month Sin	-0.269225	0.764	0.0000
Month Cos	0.176928	1.1934	0.0000
Dow Sin	0.168729	1.1839	0.0000
Dow Cos	-0.168792	0.8445	0.0647
Vehicle Tier Other	0.049580	1.0508	0.3701



Model Limitations

Limitations and Challenges

- **Outliers** were handled by regrouping - Most data was categorical.
- All Wichita data describes **interactions that ended in citations**. What other police interactions are we not capturing, in particular those associated with racial profiling (eg. searches, frisks)?
- **Officer and court/judge related data was unavailable** which may have strengthened the model
- Race is recorded at the **visual discretion** of the officer



Prediction

Predictive Performance: Out of Sample Predictions

Accuracy = 0.906

Precision = 0.906

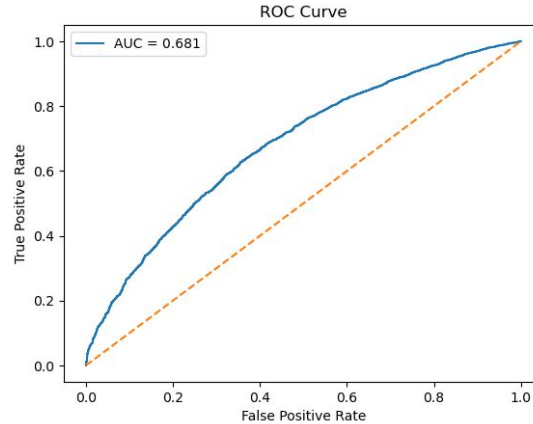
Recall (TPR) = 1.000

Specificity (TNR) = 0.000

False positive rate = 1.000

False negative rate = 0.000

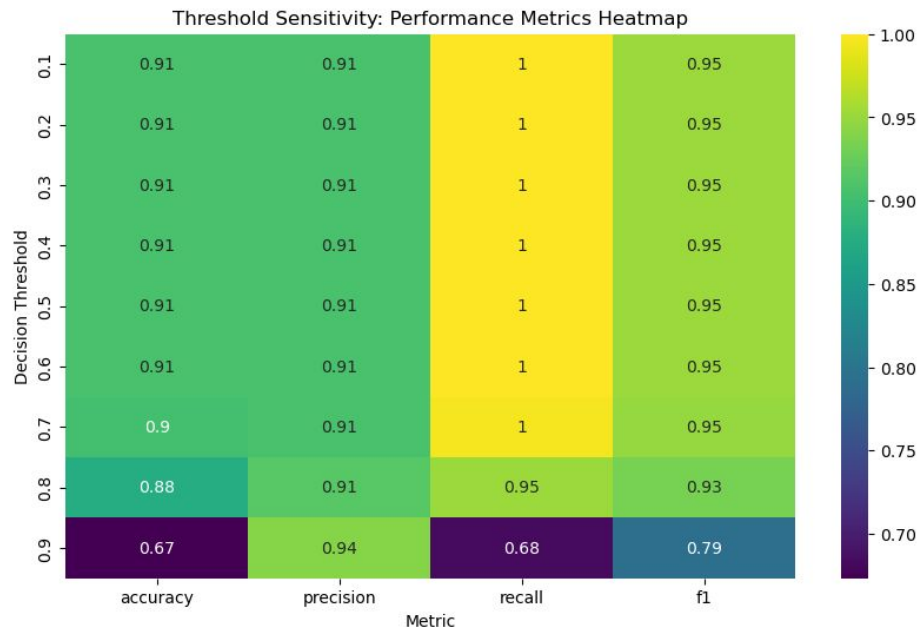
AUC = 0.681



- High recall and precision attributed to the **high skew** towards a “severe” outcome.
- The model has **weak discriminatory ability** with the AUC metric **0.681** only slightly > **0.5**. The model does not separate severe vs. non-severe cases well
- The model has a **specificity = 0.0** (the model fails to detect any non-severe cases) FPR = 1.0.
- High accuracy in the model may be misleading however **poor specificity and AUC illustrate an important finding in academic justice research**. Predictive performance does not guarantee fair decision making.

Calibrating Threshold Values for Improving Metrics

Threshold sensitivity analysis analyzes how classification metrics vary as thresholds vary. As mentioned earlier, a threshold of 0.5 produces a model with weak discrimination ability which has a **very real potential to create severe bias** in the justice system

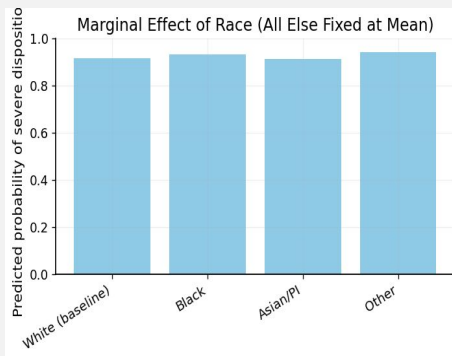


- Since the the dataset is dominated by the positive class, **0.5 may not be an appropriate threshold**
- On testing different thresholds and producing respective metrics, we can see meaningful, **unbiased results starting from 0.7**
- Recall remains **excellent while false positive rates reduce by nearly 50%**
- This is critical as high false positive rates have been linked to **highly discriminatory justice systems**
- It also contradicts legal principles prioritizing individual liberty, fairness and due process which is why threshold setting is regularly discussed in academic research in criminology

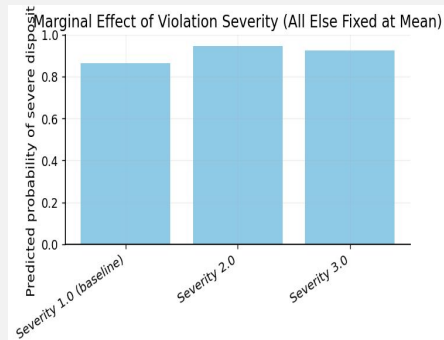


Sensitivity Analysis

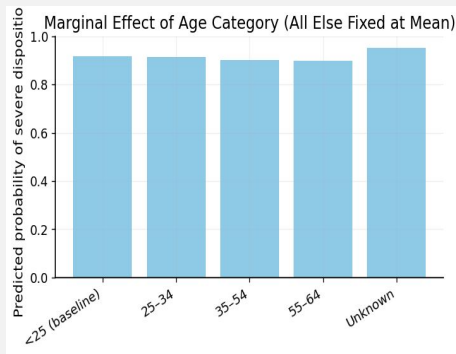
Sensitivity Analysis: Understanding Marginal Effects



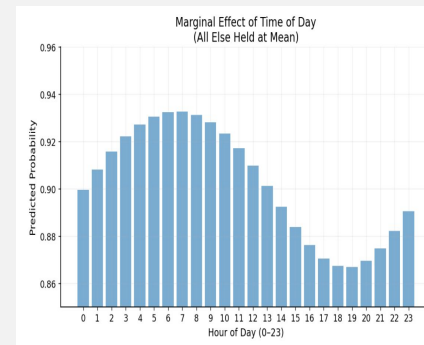
1. Black individuals have higher predicted probabilities while Asians show a lower probability
2. **Race Other** category shows the highest predicted probability
3. Overall, race has a meaningful statistical impact



1. **Violation severity** has the largest effect on the target variable
2. Interestingly, **severity 2** increases marginal probability of getting a serious disposition over severity 3
3. However, the overall structure of violation severity increasing probability of severe disposition is expected



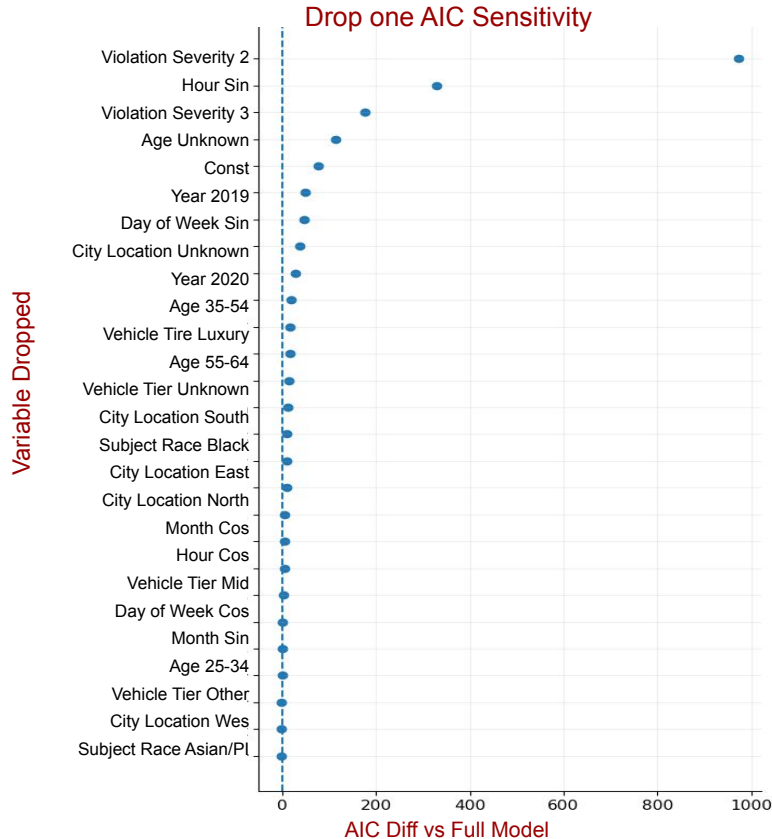
1. **Ages 35-54 and 55-64** show slightly lower probabilities compared to baseline.
2. **Unknown age** significantly increases predicted probability
3. We can see it behaves as a **data-quality** indicator correlated with severe disposition.



1. Probabilities peak between **5-7 AM** and wane between 6-8 PM
2. Time variables are statistically significant
3. This is corroborated in other statistical analyses conducted where vehicular police searches reduce in the evening as a result of the **“veil of darkness”**

Sensitivity Analysis: Effects of Variable Dropping

We assess how important each predictor is to the final model by dropping variables one by one and computing AIC. If AIC increases, it implies that the model's performance is worsening, and vice versa



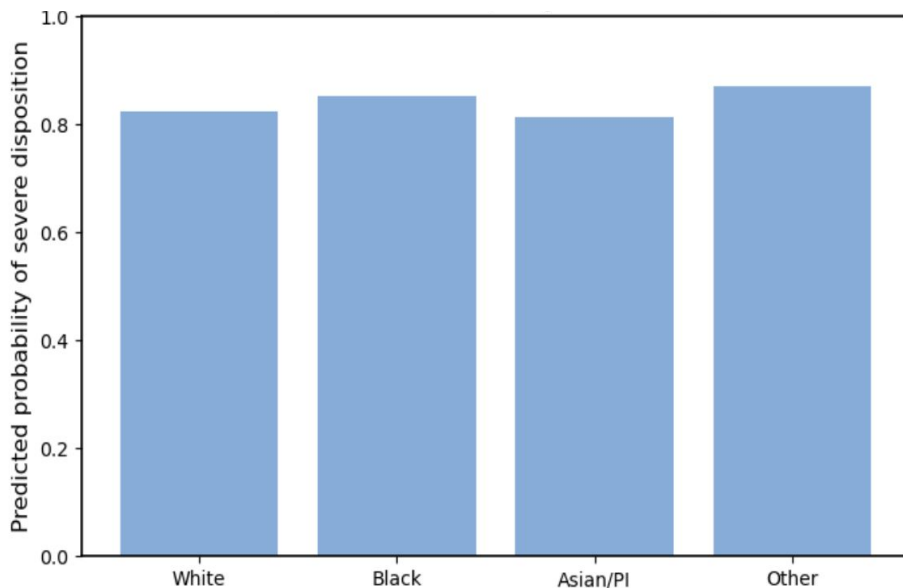
Training Data Set AIC : 38726.55

- **Violation severity** is the most insignificant predictor
- Dropping violation severity 2 increased AIC the most while dropping violation severity 3 was not far behind.
- Dropping the geometric transformations of **time variables** also increased the AIC significantly indicating that cyclical pattern of law enforcement behaviour is meaningful.
- Dropping **location variables** have a modest incremental effect on AIC as well.
- Interestingly, Dropping **age unknown** noticeably increases AIC, which may reflect an issue in the data collection process.
- While dropping race variables had very small impacts on AIC, it must be noted that they still have some strong positive coefficients

Scenario Analysis: Combining Multiple Variables

Scenario analysis helps understand how predicted disposition severity changes under different realistic combinations of conditions. Here we have used a common low risk scenario and toggled race variables

Scenario-Based Marginal Effect of Race
(low-risk context where only race varies)



Low Risk Baseline Scenario

Violation severity = low

Age 35-54

Vehicle tier = Mid

City Location = East

Time of Day = 12:00 Noon

All other dummies 0 but race varied

- In the baseline low risk scenario varied only by race, **black and other groups** still have a higher predicted probability of getting a severe disposition
- The differences are modest, but support the conclusions from the coefficient analysis
- Even small probability differences can translate into **large disparities** in false-positive rates, which can have serious real life impacts



Conclusion

Conclusions

- Perfect recipe for how to have a bad day in Wichita is: Being either a Black driver or driver of visually **indeterminate race** driving in **demographically diverse parts** of the city at around **6:41 am on a Saturday** morning around **Christmas time** and for whatever reason, the officer does not take down your correct **age** from your DL
- We expected **race to have a bigger beta** (like the literature suggests). But...what is hiding in those unknown categories? Future work should focus unpacking the “**unknowns**”
- Data can lead to policy changes but also **policy can shape data**. When you are interacting with a dataset, you are also interacting with the people and pressures who created it
- We can't believe that **race is visually assigned** by authority. Possible extension would be to chase down “race corrections” in the citation paperwork.

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Appendix

Generalized Linear Model Regression Results

```

=====
Dep. Variable:    disposition_severity    No. Observations:    86830
Model:            GLM                    Df Residuals:        86803
Model Family:     Binomial                Df Model:            26
Link Function:    Logit                   Scale:              1.0000
Method:           IRLS                    Log-Likelihood:      -25787.
Date:             Sat, 06 Dec 2025         Deviance:            51574.
Time:             12:57:20                 Pearson chi2:        8.65e+04
No. Iterations:   6                       Pseudo R-squ. (CS): 0.03046
Covariance Type:  nonrobust
=====

```

	coef	std err	z	P> z	[0.025	0.975]
const	1.7385	0.038	46.013	0.000	1.664	1.813
subject_race_black	0.1671	0.049	3.391	0.001	0.070	0.264
subject_race_asian/pacific islander	-0.1771	0.092	-1.935	0.053	-0.357	0.002
subject_race_other	0.4941	0.372	1.329	0.184	-0.235	1.223
vehicle_tier_LUXURY	-0.2621	0.053	-4.968	0.000	-0.366	-0.159
vehicle_tier_MID	-0.1038	0.048	-2.172	0.030	-0.197	-0.010
vehicle_tier_UNKNOWN	-0.2525	0.054	-4.696	0.000	-0.358	-0.147
violation_severity_2.0	1.0300	0.029	35.561	0.000	0.973	1.087
violation_severity_3.0	0.6055	0.042	14.288	0.000	0.522	0.689
city_location_east	-0.1202	0.037	-3.228	0.001	-0.193	-0.047
city_location_north	-0.2589	0.066	-3.898	0.000	-0.389	-0.129
city_location_south	-0.1620	0.042	-3.816	0.000	-0.245	-0.079
city_location_unknown	-0.2823	0.038	-7.506	0.000	-0.356	-0.209
city_location_west	-0.0540	0.044	-1.235	0.217	-0.140	0.032
age_category_25-34	-0.0392	0.034	-1.150	0.250	-0.106	0.028
age_category_35-54	-0.1730	0.031	-5.499	0.000	-0.235	-0.111
age_category_55-64	-0.2277	0.043	-5.241	0.000	-0.313	-0.143
age_category_Unknown	0.5823	0.047	12.411	0.000	0.490	0.674
year_cat_2019	0.1736	0.026	6.612	0.000	0.122	0.225
year_cat_2020	0.6097	0.048	12.582	0.000	0.515	0.705
hour_sin	0.4060	0.017	23.249	0.000	0.372	0.440
hour_cos	-0.0732	0.020	-3.697	0.000	-0.112	-0.034
month_sin	-0.2692	0.017	-15.539	0.000	-0.303	-0.235
month_cos	-0.0381	0.018	-2.162	0.031	-0.073	-0.004
dow_sin	-0.1688	0.019	-8.976	0.000	-0.206	-0.132
dow_cos	-0.0310	0.017	-1.847	0.065	-0.064	0.002
vehicle_tier_other	0.0496	0.055	0.896	0.370	-0.059	0.158

Appendix – Threshold sensitivity analysis table

	threshold	accuracy	precision	recall	f1	TN	FP	FN	TP
0	0.1	0.905657	0.905657	1.000000	0.950493	0	2048	0	19660
1	0.2	0.905657	0.905657	1.000000	0.950493	0	2048	0	19660
2	0.3	0.905657	0.905657	1.000000	0.950493	0	2048	0	19660
3	0.4	0.905657	0.905657	1.000000	0.950493	0	2048	0	19660
4	0.5	0.905657	0.905657	1.000000	0.950493	0	2048	0	19660
5	0.6	0.905703	0.905699	1.000000	0.950516	1	2047	0	19660
6	0.7	0.902985	0.906719	0.995270	0.948933	35	2013	93	19567
7	0.8	0.875438	0.914572	0.951322	0.932585	301	1747	957	18703
8	0.9	0.673346	0.940430	0.682553	0.791005	1198	850	6241	13419

Appendix – Variable sensitivity analysis figures

Race Category	Predicted Severe Probability
White (baseline)	0.916028
Black	0.929815
Asian/PI	0.911292
Other	0.939962

Age Category	Predicted Severe Probability
25–34	0.914714
35–54	0.907546
55–64	0.901197
Unknown	0.947240

Violation Severity	Predicted Severe Probability
Severity 2.0	0.949559
Severity 3.0	0.951637

Hour of Day	Predicted Severe Probability
0	0.899354
1	0.907932
2	0.915641
3	0.922102
4	0.927093
5	0.930506
6	0.932303
7	0.932474
8	0.931018
9	0.927942
10	0.923276
11	0.917110
12	0.909637
13	0.901201
14	0.892328
15	0.883708
16	0.876145
17	0.870433
18	0.867227
19	0.866915
20	0.869534
21	0.874767
22	0.882006
23	0.890472

Appendix – Variable drop sensitivity figures

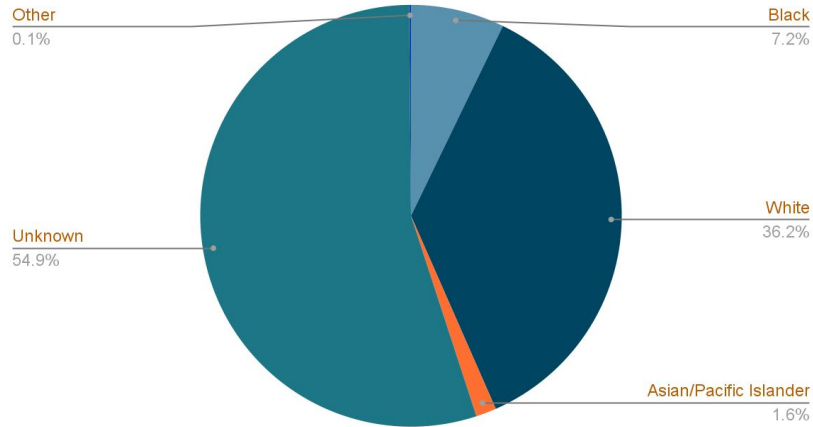
Variable	AIC_dropped	AIC_diff
subject_race_asian/pacific islander	38725.000372	-1.546810
city_location_west	38725.057514	-1.489667
subject_race_other	38725.378565	-1.168616
vehicle_tier_other	38725.481216	-1.065966
age_category_25-34	38725.492704	-1.054478
month_sin	38726.102549	-0.444633
dow_cos	38728.506807	1.959625
vehicle_tier_MID	38730.309609	3.762427
hour_cos	38731.283742	4.736560
month_cos	38731.973772	5.426590
city_location_north	38734.749067	8.201885
city_location_east	38736.473638	9.926456
subject_race_black	38736.619965	10.072783
city_location_south	38738.950333	12.403151
vehicle_tier_UNKNOWN	38740.238963	13.691781
age_category_55-64	38741.777729	15.230547
vehicle_tier_LUXURY	38742.202138	15.654956
age_category_35-54	38744.014751	17.467569
year_cat_2020	38753.356944	26.809762
city_location_unknown	38762.701645	36.154463
dow_sin	38773.181212	46.634030
year_cat_2019	38775.100392	48.553210
const	38803.709872	77.162690
age_category_Unknown	38838.977729	112.430547
violation_severity_3.0	38901.839530	175.292348
hour_sin	39056.153289	329.606107
violation_severity_2.0	39699.432066	972.884884

Future Improvements

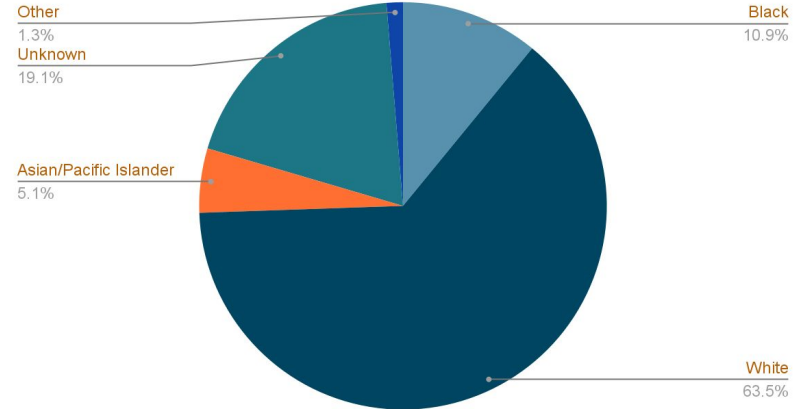
- We need to further clarify the **demographics** of Wichita and their relationship to citations
- Refine the **violation classification** further based on Kansas traffic law
- Look into **race unknown and sex unknown** (dropped due to VIF). Are these categories masking racial and gender diverse publics?
- **Rebalance** the data, weighted loss or resampling
- Explore **ML tools** like LIME and SHAP
- Evaluate **group fairness metrics** (FPR disparity, calibration by group)
- Model accuracy alone is not meaningful in imbalance justice datasets
- Threshold choice is a fairness decision, not a technical one
- Race still shows **statistically significant relationships**, so oversight and fairness evaluation are essential

Chi-Square Statistic Reflects Racial Discrepancies

Citations By Race



Racial Demographics of Wichita



Chi-square Goodness of Fit Statistic: 72321.49
p-value: 0.0